TrES-1b

R-0103

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-1_260425` · created 2026-07-16T03:17:58+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461155.8913 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1845 +/- 0.0085
Transit depth (Rp/Rs)^2	3.41 +/- 0.31 [%]
Orbital Inclination (inc)	87.0 +/- 1.03
Airmass coefficient 1 (a1)	0.962 +/- 0.021
Airmass coefficient 2 (a2)	0.036 +/- 0.022
Scatter in the residuals of the lightcurve fit is	2.2 %
Best Comparison Star	#2 - [463, 158]
Optimal Aperture	3.29
Optimal Annulus	10.24
Transit Duration (day)	0.0941 +/- 0.0098

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1845 ± 0.0085 vs 0.1358 (deviation 5.73σ)
Duration fit vs published	0.0941 d vs 0.1042 d (deviation 1.03σ)
Mid-time O–C	0.9 min
Depth SNR	21.7
Residual scatter	2.2 %

Light curve

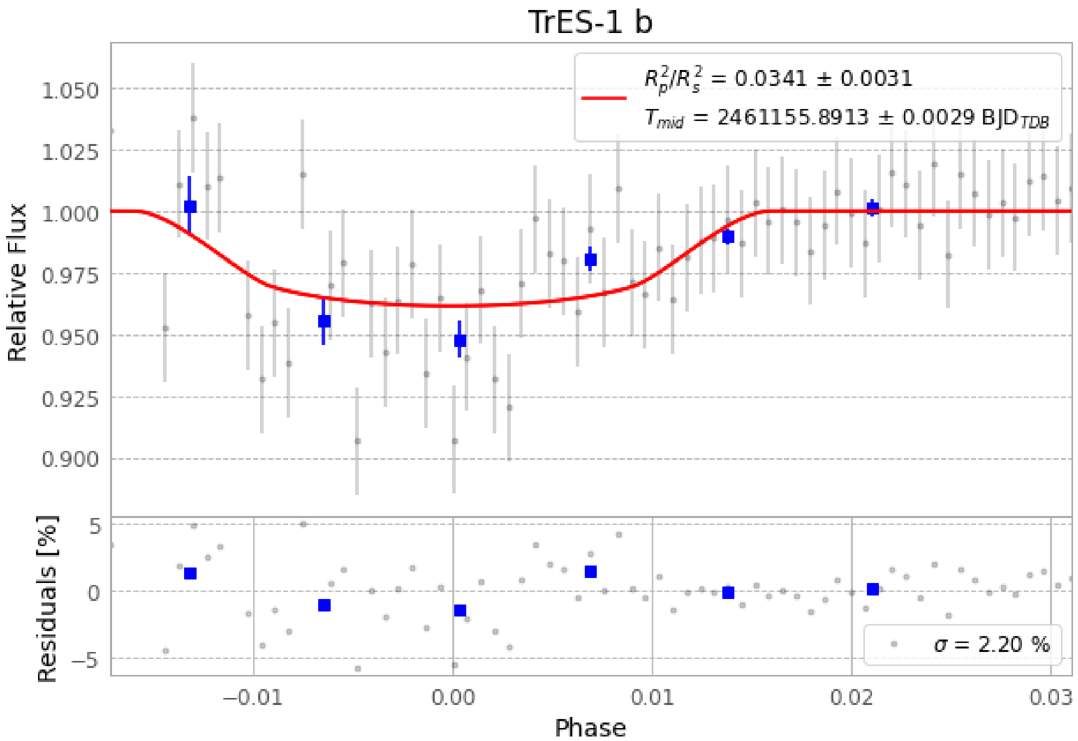

Figure 1 — FinalLightCurve_TrES-1 b_2026-04-25.png (SHA-256 5971487ce628622e…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [256, 254], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 39f3a8c9d08e137f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

72 FITS frames (47 MB), `TRES-1260425080315.FITS` → `TRES-1260425113615.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-260425.log (003d4d17426f…), FinalLightCurve_TrES-1 b_2026-04-25.png (5971487ce628…), FinalLightCurve_TrES-1 b_2026-04-25.csv (d23c6f3bf214…)

Compute resources

Wall time 190.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 03:17Z.